FERROUS SULFATE

Empty stomach (if tolerated)

Fatal overdose in children

CHEMICAL NAME Iron(II) sulfate ($FeSO_4 \cdot 7H_2O$ or dried $FeSO_4$); 200 mg dried ferrous sulfate ≈ 65 mg elemental iron

CLASS Oral iron supplement

TRADE NAMES Feospan · Fer-In-Sol (drops) · Ferrograd · Slow-Fe

THERAPEUTIC USES / INDICATIONS

- Iron-deficiency anaemia (treatment).
- Prevention of iron deficiency: pregnancy, infancy, menorrhagia, after gastrectomy, CKD (oral trial).
- Blood donors; preterm infants.

ADMINISTRATION

- **PO** ideally 1 h before meals with water or orange juice (vitamin C ↑ absorption); with food if GI upset.
- Avoid tea, coffee, milk, antacids, calcium within 2 h.
- Liquid preparations stain teeth — use straw, rinse mouth.
- Alternate-day dosing improves absorption and tolerance.

DOSAGES

- **Treatment (adult):** 200 mg once daily (65 mg elemental iron) — once daily or alternate days.
- **Prophylaxis:** 200 mg once daily.
- **Children:** 3 mg/kg/day elemental iron (max 200 mg/day).
- Continue 3 months after Hb normalises to replenish stores.

PHARMACOKINETICS

- Absorbed in duodenum/jejunum (Fe^{2+}), ↑ in deficiency; regulated by hepcidin.
- Stored as ferritin/haemosiderin; not actively excreted (lost via GI shedding, menstruation).

PHARMACODYNAMICS

- Provides **iron for haemoglobin**, myoglobin and enzyme synthesis → reticulocyte response in 5–10 days; Hb rises ~10–20 g/L every 2–3 weeks.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, epigastric pain, constipation or diarrhoea.
- Black stools (harmless).
- Teeth staining (liquid).

ADVERSE EFFECTS (SERIOUS)

- Iron overdose in children: vomiting, GI haemorrhage, metabolic acidosis, liver failure (antidote: desferrioxamine).
- Iron overload with long-term use without deficiency.

CONTRAINDICATIONS & CAUTIONS

- Haemochromatosis, haemosiderosis; iron overload.
- Anaemia not due to iron deficiency (e.g. thalassaemia without deficiency).
- Repeated blood transfusions.

KEY INTERACTIONS

- **Tetracyclines, fluoroquinolones, levothyroxine, bisphosphonates, levodopa, mycophenolate:** ↓ absorption (separate by 2–4 h).
- **PPIs, antacids, calcium, tea:** ↓ iron absorption.

NURSING MANAGEMENT

- Check Hb, ferritin; recheck Hb at 2–4 weeks.
- Monitor bowel habit; manage constipation.
- Investigate cause of iron deficiency (GI blood loss in men and postmenopausal women).
- Store safely out of children's reach.

PATIENT EDUCATION

- Take on an empty stomach with orange juice; with food if it upsets your stomach.
- Stools will be black — this is normal; report black tarry stools with dizziness.
- Keep tablets locked away; iron overdose is dangerous for children.
- Eat iron-rich foods (meat, beans, green vegetables).

KEY POINTS

- Oral iron — 200 mg (65 mg Fe) daily.
- GI upset, black stools.
- Continue 3 months after Hb normal.
- Child overdose dangerous; many absorption interactions.